

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Computer Science and Engineering)(CS201)
Master Subject Coordinator Name:	Munish Kumar	Master Subject Coordinator E-Code:	E16513
Course Name	Professional ethics and human values	Course Code	22CST-451

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
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Course Type	Course Category	Mode of Assessment	Mode of Delivery
N.A	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1: To provide practical knowledge using state-of-the-art technological support for the experiential learning of our students. M2: To provide industry recommended curriculum and transparent assessment for quality learning experiences. M3: To create global linkages for interdisciplinary collaborative learning and research. M4: To nurture advanced learning platform for research and innovation for students' profound future growth. M5: To inculcate leadership qualities and strong ethical values through value based education.
Vision of the Department	To be recognized as a leading Computer Science and Engineering department through effective teaching practices and excellence in research and innovation for creating competent professionals with ethics, values and entrepreneurial attitude to deliver service to society and to meet the current industry standards at the global level.

Program Educational Objectives(PEOs)

PEO1	Engage in successful careers in industry, academia, and public service, by applying the acquired knowledge of Science, Mathematics and Engineering, providing technical leadership for their business, profession and community.
PEO2	Establish themselves as entrepreneur, work in research and development organization and pursue higher education.
PEO3	Exhibit commitment and engage in lifelong learning for enhancing their professional and personal capabilities.

Program Specific OutComes(PSOs)

PSO1	Exhibit attitude for continuous learning and deliver efficient solutions for emerging challenges in the computation domain.
PSO2	Apply standard software engineering principles to develop viable solutions for Information Technology Enabled Services (ITES).

Program OutComes(POs)

PO1	Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals and computer science fundamental and strategies which have the solution of complex computer science engineering problems.
PO2	Problem analysis: Identify, formulate, research literature, and analyze complex computer science engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex database and software engineering problems and design system components or processes that meet the specified needs with appropriate considerations for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of software engineering & networking based experiments, analysis and Interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern Computer science engineering and IT tools including prediction and modeling to complex database or software engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess Social, health, safety, legal and cultural issues and the consequent responsibilities relevant to the Professional Computer Science & Engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional computer science and engineering solutions in social and environmental contexts, and demonstrate the knowledge of, and need for sustainable development goals.

PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of computer science engineering practice
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex computer science engineering activities with the engineering community like CSI society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the computer science engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context to technological change.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	A Foundation Course in Human Values and Professional Ethics	RR Gaur, R Sangal, G P Bagaria,	First	Excel Books	2009

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Energy & Equity	Ivan Illich	I	The Trinity Press,	1974
2	How the Other Half Dies	Sussan George	I	Penguin Press	1976
3	Small is Beautiful: a study of economics as if people mattered	E.F. Schumacher	I	RHUK	1973

Course OutCome	
SrNo	OutCome
CO1	Demonstrate the basic significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content, and process of value
CO2	Apply the meaning of Harmony in the Self the Co-existence of Self and Body and Distinguish between the Self and the Body
CO3	Analyze the value of harmonious relationships based on trust, respect, and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society.
CO4	Interpret the interconnectedness of nature and existence to align human participation with natural systems.
CO5	Propose evidence-based solutions to real-world problems and formulate actionable plans to maintain environmental and social harmony.

Lecture Plan Preview-Theory							
Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Number(s)	BT Level
1	1	Human Values	Introduction to Human Values; Meaning and significance of Morals, Values, and Ethics; Importance of value education in shaping responsible personal and professional behaviour.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Case Study,PPT,V ideo Lecture	CO1	BT1
1	2	Human Values	Integrity and Work Ethics; Honesty in behavior; Professional responsibility and accountability; Role of ethical conduct in workplace harmony and productivity.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,PPT	CO1	BT2

1	3	Human Values	Service-Learning and Civic Virtue; Community participation; Social responsibility; Respect for others; Role of civic sense in maintaining social harmony.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,Video Lecture	CO1	BT2
1	4	Human Values	Living Peacefully; Caring and Sharing; Mutual respect; Cooperative living; Importance of empathy and kindness for building healthy relationships.	T-A Foundation Course in Human V	Activity,Case Study,PPT	CO1	BT2
1	5	Human Values	Honesty, Courage, Valuing Time, Cooperation; Importance of truthful conduct; Decision-making with courage; Time management; Teamwork; Introduction to Empathy, Self-Confidence, Character, and Spirituality.	T-A Foundation Course in Human V	Activity,Case Study,PPT	CO1	BT2
1	6	Engineering Ethics	Introduction to Senses of Engineering Ethics; Meaning, scope, and importance of ethics in engineering practice; Distinction between personal and professional ethics.	T-A Foundation Course in Human V	Case Study,PPT,Video Lecture	CO1	BT2
1	7	Engineering Ethics	Variety of Moral Issues in engineering; Types of ethical inquiry—normative, descriptive, and meta-ethics; Introduction to Moral Dilemmas with real-life engineering examples.	T-A Foundation Course in Human V	Activity,Case Study,PPT	CO1	BT2
1	8	Engineering Ethics	Moral Autonomy: Concept, characteristics, and significance; Kohlberg's Theory of Moral Development—Stages and engineering relevance.	T-A Foundation Course in Human V	Infographics, PPT,Video Lecture	CO1	BT2
1	9	Engineering Ethics	Gilligan's Theory of Moral Development; Consensus and Controversy in engineering decision-making; Balancing stakeholder views in ethical issues.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,PPT	CO1	BT3
1	10	Engineering Ethics	Professions and Professionalism; Professional ideals and virtues; Theories about right action—self-interest, customs, religion; Uses and applications of ethical theories in engineering.	T-A Foundation Course in Human V	Case Study,PPT,Video Lecture	CO1	BT3
2	11	Engineering As Social Experimentation	Concept of Engineering as Social Experimentation; Engineering decisions as trial-action processes; Similarities between engineering projects and scientific experiments.	T-A Foundation Course in Human V	Infographics, PPT,Video Lecture	CO2	BT2
2	12	Engineering As Social Experimentation	Engineers as Responsible Experimenters; Conscientiousness, accountability, and moral responsibility; Ethical obligations during design, testing, and implementation.	T-A Foundation Course in Human V	Activity,Case Study,PPT	CO2	BT3
2	13	Engineering As Social Experimentation	Codes of Ethics: Purpose, importance, structure; Overview of engineering codes (ASME, IEEE, IEI); Role of codes in guiding ethical decisions.	T-A Foundation Course in Human V	Infographics, PPT,Simulation	CO2	BT2

2	14	Engineering As Social Experimentation	Balanced Outlook on Law; Relationship between ethics and law; When law supports ethics and when it may conflict; Legal vs. moral responsibilities of engineers.	T-A Foundation Course in Human V	Case Study, Instructor Lead Workshop, PPT	CO2	BT4
2	15	Engineering As Social Experimentation	Case Study—Columbia Disaster	T-A Foundation Course in Human V	Activity, Case Study, Reports, Video Lecture	CO2	BT4
2	16	Safety, Responsibilities, And Rights	Concept of Safety and Risk in engineering; Types of risks; Distinction between acceptable and unacceptable risk; Role of engineers in ensuring safety.	T-A Foundation Course in Human V	Infographics, PPT, Video Lecture	CO2	BT2
2	17	Safety, Responsibilities, And Rights	Assessment of Safety and Risk; Quantitative and qualitative risk analysis; Risk-benefit evaluation; Techniques for reducing risk in engineering design and projects.	T-A Foundation Course in Human V	Case Study, PPT, Simulation	CO2	BT3
2	18	Safety, Responsibilities, And Rights	Collegiality, Loyalty, and Respect for Authority; Professional teamwork; Ethical collaboration; Conflicts between loyalty and professional judgment.	T-A Foundation Course in Human V	Activity, Flipped Classes, PPT	CO2	BT2
2	19	Safety, Responsibilities, And Rights	Confidentiality, Conflicts of Interest, Occupational Crime, Employee Rights, Professional Rights, Intellectual Property Rights (IPR), Discrimination in the workplace.	T-A Foundation Course in Human V	Case Study, PPT, Reports	CO2	BT4
2	20	Safety, Responsibilities, And Rights	Titanic Disaster: Engineering failures, safety negligence, risk miscalculations, communication failures, ethical lapses, lessons on responsibility and safety culture.	T-A Foundation Course in Human V, R-Energy & Equity, R-How the Other Half Dies, R-Small is Beautiful: a study of	Activity, Case Study, Reports, Video Lecture	CO2	BT4
3	21	Global Issues	Multinational Corporations (MNCs); Globalization and ethical responsibilities; Ethical challenges in cross-cultural environments; Corporate accountability and social responsibility.	T-A Foundation Course in Human V	Infographics, PPT, Video Lecture	CO3	BT2
3	22	Global Issues	Environmental Ethics; Sustainable development; Environmental responsibilities of engineers; Pollution, resource depletion, climate ethics and green engineering practices.	T-A Foundation Course in Human V, R-Energy & Equity, R-How the Other Half Dies, R-Small is Beautiful: a study of	Activity, Case Study, PPT	CO3	BT3
3	23	Global Issues	Computer Ethics; Privacy, security, data protection, intellectual property in digital space; Ethical issues in AI, automation, surveillance and cybersecurity.	T-A Foundation Course in Human V	PPT, Reports, Simulation	CO3	BT4
3	24	Global Issues	Weapons Development; Ethical concerns in military technologies; Dual-use technologies; Role of engineers in national security; Peace engineering.	T-A Foundation Course in Human V	Case Study, PPT, Video Lecture	CO3	BT4
3	25	Global Issues	Engineers as Managers, Consulting Engineers, Expert Witnesses and Advisors; Professional judgment; Communication responsibilities; Ethical decision-making in leadership roles.	T-A Foundation Course in Human V	Activity, Instructor Lead Workshop, PPT	CO3	BT3

3	26	Moral Leadership	Concept of Moral Leadership; Ethical decision-making; Traits of ethical leaders; Role of engineers in promoting ethical culture in organizations.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	PPT	CO1	BT2
3	27	Moral Leadership	ASME & Environmental Responsibilities: interpret how ASME code clauses relate to safety, sustainability, and protection of natural systems; work with examples from energy and manufacturing projects.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	PPT	CO4	BT3
3	28	Moral Leadership	ASCE (American Society of Civil Engineers) Code of Ethics: Fundamental principles, duties of engineers, public safety, conflicts of interest, and integrity standards.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3	BT3
3	29	Moral Leadership	IEI (Institution of Engineers India): Responsibilities to society, employer, and profession. IIMM: Ethical procurement and materials management. IETE: Technical competence, confidentiality, safety, and integrity.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3	BT3
3	30	Moral Leadership	Comparison of codes of ethics across engineering bodies (ASME, ASCE, IEEE, IEI, IIMM, IETE); Common ethical principles; How codes guide real-world ethical dilemmas; Unit-3 assessment and discussion.	,T-A Foundation Course in Human V,R-Energy & Equity,R-How the Other Half Dies,R-Small is Beautiful: a study of	Activity,Case Study,PPT,R eports	CO5	BT4

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	External Theory	60	60
2	Assignment/PBL	10	10
3	Attendance Marks	2	2
4	Mid-Semester Test-1	20	10
5	Quiz	4	4

6	Surprise Test	12	4
7	Mid-Semester Test-2	20	10

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CO2	3	3	NA	3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CO3	NA	3	2	3	3	NA	NA	NA	NA	NA	NA	NA	NA	NA
CO4	3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CO5	NA	3	2	3	3	NA	NA	NA	NA	NA	NA	1	NA	2
Target	3	3	2	3	3	NA	NA	NA	NA	NA	NA	1	NA	2

